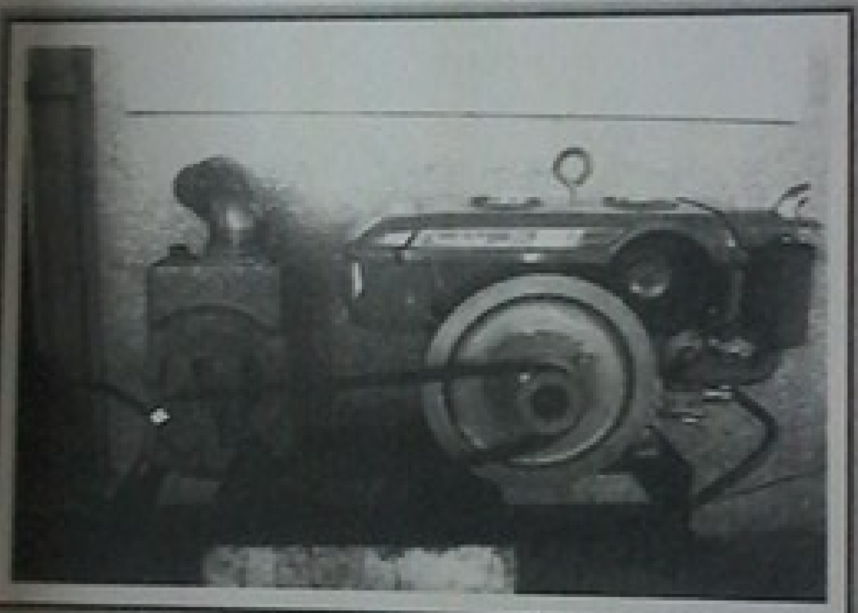
AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2014 - 13



KITACO NS-100 Centrifugal Pump coupled to GOLDEN BOW GP95 Diesel Engine

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Test Report No.	:	2014 - 13
Date of Test	:	February 16, 2014
Test Valid Until	:	February 16, 2019
Brand	:	KITACO
Model	:	NS-100
Serial No.	:	No Data
Type and Size	:	Single-end suction, 100 x 100 mm, Non-Self Priming Centrifugal pump
Manufacturer	:	No Data
Test Requested by	:	M&W DISTRIBUTING CO., INC. 14 Panay Avenue, Quezon City Sta. Cruz, Manila
Machine Classification	:	Commercial

SUMMARY

The KITACO NS-100 centrifugal pump coupled to a GOLDEN BOW GP95 diesel engine was tested for performance and suction condition. The pump is a 100 x 100 mm non self-priming centrifugal pump. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with actual diameter of 102 mm (4.0 inches) and 114.3 mm (4.5 inches), respectively and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 23.6 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2273 and 2570 rpm, respectively with Engine Fuel Consumption Rate of 2.33 L/h. The Maximum Discharge Capacity of 20.5 L/s occurred at 5.02 m Total Head with Maximum Fuel Consumption Rate of 3.79 L/h. The shaft speed of the pump and engine at this point were 2003 and 2359 rpm respectively.

The Maximum System Efficiency of 5.25% occurred at 11.3 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 8.85 m. The Net Positive Suction Head Available at this point was 1.03 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1.**

The test was conducted on February 16, 2014 at the AMTEC Test Laboratory, UPLB, College, Laguna.

Figure 1. KITACO NS-100 Centrifugal pump coupled to GOLDEN BOW GP95 Diesel engine on the test set-up

METHOD OF TEST

The pump was tested following the "Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test".

DESCRIPTION OF THE PUMP

The KITACO NS-100 Centrifugal pump coupled to GOLDEN BOW GP95 Diesel Engine shown in **Figure 2**, was a single-end suction, 100 mm x 100 mm, centrifugal pump coupled to 6.99 kW (9.37 Hp) GOLDEN BOW GP95 diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameter of the engine and pump pulleys were 102 mm (4.0 inches) and 114.3 mm (4.5 inches), respectively. The suction inlet and discharge outlet were threaded and designed for a 100 mm diameter pipe. The impeller was a semi-open type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a packing seal. The pump shaft rotated on ball bearing kept in place by a bearing housing integral to the casing.

The prime mover, a 6.99 kW GOLDEN BOW GP95 diesel engine was a four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized an oil-bath type air cleaner. Lubrication system was force-feed lubricating type. Cooling was done by water circulating from cylinder jacket to the radiator using thermo-siphon principle. A fan driven by the pulley on the flywheel was used to cool the warm water in the radiator. It was started using manual cranking lever aided by a manually actuated decompression lever. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.

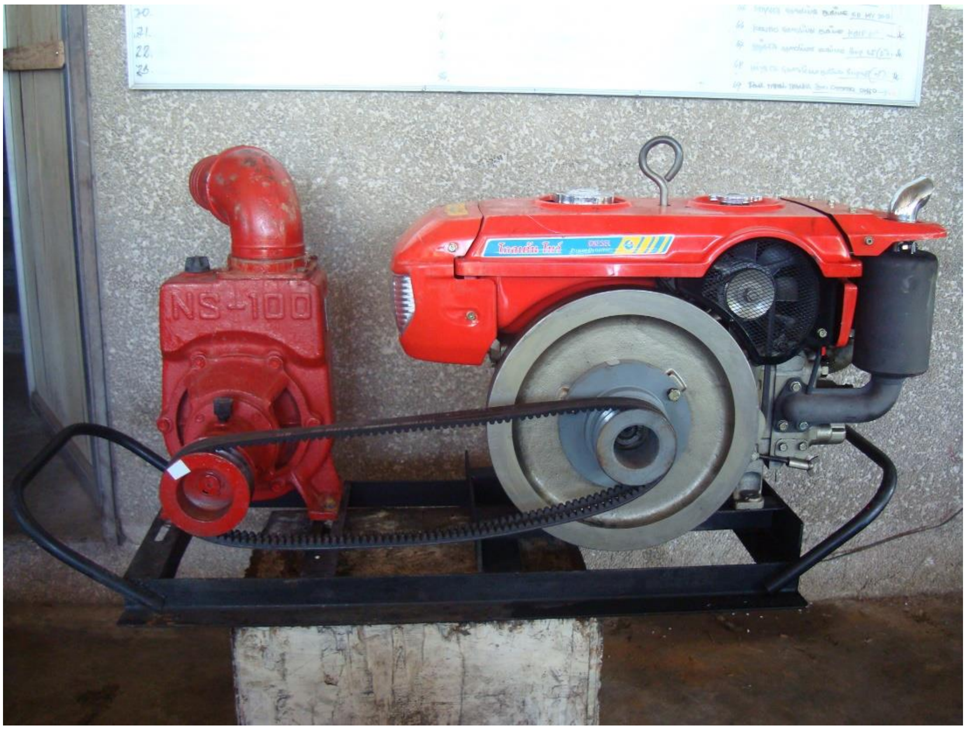

Figure 2. KITACO NS-100 Centrifugal Pump coupled to a GOLDEN BOW GP95 Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KITACO NS-100 Centrifugal Pump coupled to GOLDEN BOW GP95 Diesel Engine

1.	Brand		:	KITACO
2.	Model		:	NS-100
3.	Serial No.		:	No Data
4.	Make		:	No Data
5.	Type		:	Single-end suction, Centrifugal pump
6.	Size (mm)		:	100 x 100
7.	Overall Dimension and Weight (Including mounting frame and engine)			
	7.1	Length (mm)	:	1320
	7.2	Width (mm)	:	485
	7.3	Height (mm)	:	600
	7.4	Weight (kg)	:	176
8.	Impeller			
	8.1	Type	:	Semi-open
	8.2	Diameter (mm)	:	195
	8.3	Width (mm)	:	42
	8.4	No. of vanes	:	5
	8.5	Material	:	Brass
9.	Suction Inlet			
	9.1	Type	:	Threaded flange
	9.2	Diameter (mm)	:	100
	9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
	10.1 Type	:	Threaded flange
	10.2 Diameter (mm)	:	100
	10.3 Material	:	Cast Iron
11.	Casing		
	11.1 Type	:	Volute
	11.2 Material	:	Cast Iron
12.	Stuffing box	:	Packing Seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	2000
15.	Rated Maximum Discharge Capacity (L/s)	:	28.3
16.	Rated Maximum Total Head (m)	:	25
17.	Primemover (based on Plate Rating)		
	17.1 Brand	:	GOLDEN BOW
	17.2 Model	:	GP95
	17.3 Serial No.	:	G95-11112389
	17.4 Make	:	CHONGQING GOLDEN BOW GROUP POWER CO., LTD.
	17.5 Type	:	Single cylinder, 4-stroke, water-cooled, horizontal cylinder diesel engine
	17.6 Rated Output Power (kW)		
	Maximum	:	6.99 @ 2400 rpm
	Continuous	:	6.25 @ 2400 rpm

Table 1. (Cont'n)

17.8	Bore x Stroke (mm)	:	No Data
17.9	Displacement Volume (cc)	:	522
17.10	Cooling System	:	Water-cooled
17.11	Lubricating System	:	Force-feed
17.12	Starting System	:	Rope-recoil
17.13	Air Cleaner	:	Oil Bath type
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pump Tested	:	KITACO NS-100 coupled to GOLDEN BOW GP95 diesel engine
Date of Test	:	February 16, 2014
Test Condition		
Ambient Temperature (^{o}C)	:	25.9
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (^{o}C)	:	26.6

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP-TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2570	2273	2.33	23.6	0	0	24.2	0
2567	2245	2.34	21.4	4.38	0.98	24.4	3.76
2556	2231	2.50	19.6	5.85	1.12	26.0	4.30
2548	2215	2.65	17.6	8.31	1.43	27.6	5.19
2545	2200	2.69	15.5	9.36	1.42	28.0	5.08
2526	2177	2.73	13.5	11.3	1.49	28.4	5.25
2517	2161	2.72	12.1	12.6	1.49	28.3	5.24
2503	2151	2.79	10.7	13.4	1.40	29.1	4.82
2487	2129	2.85	8.79	14.5	1.25	29.6	4.22
2359	2003	3.79	5.02	20.5	1.01	39.5	2.56

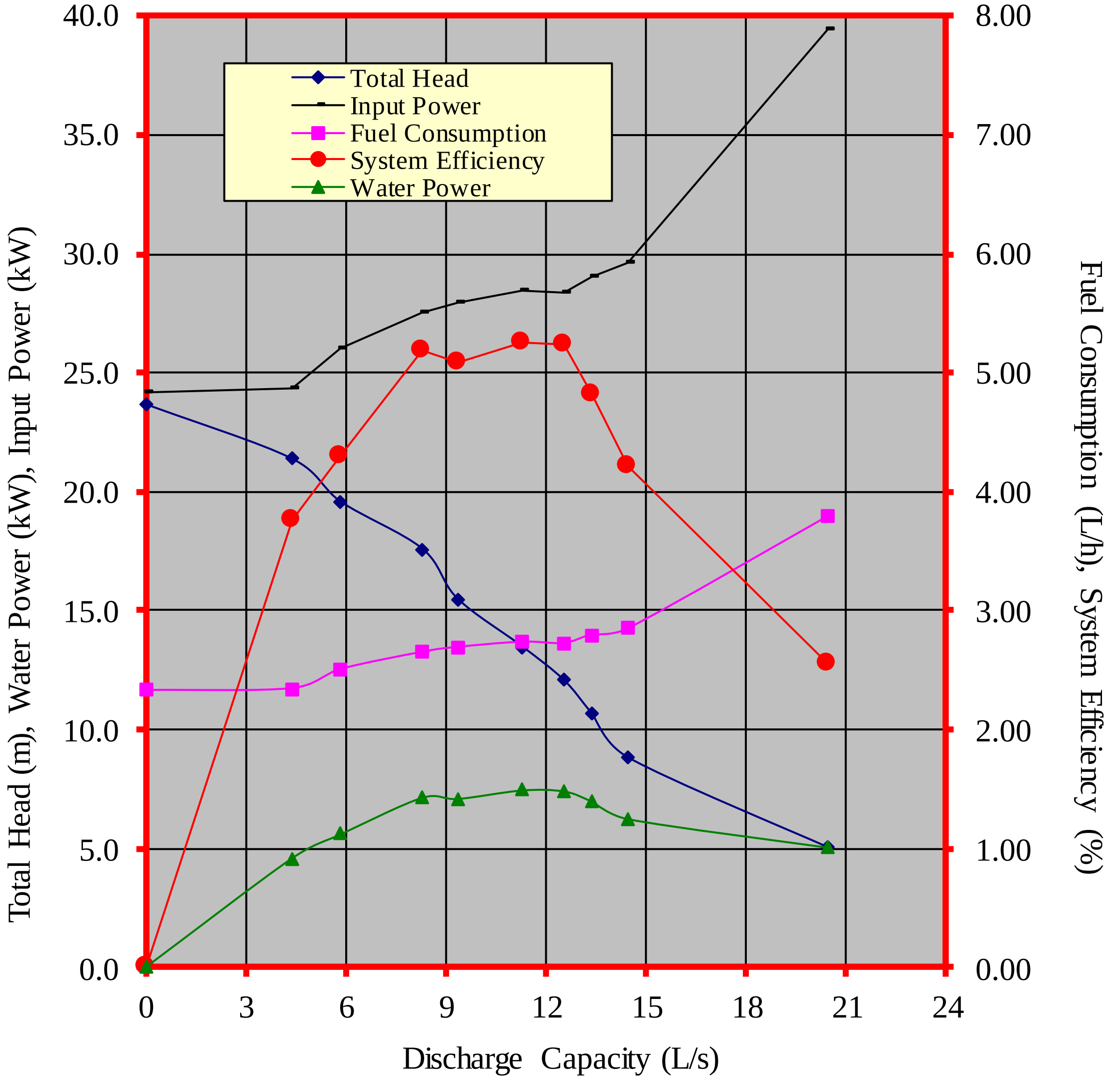

Figure 3. Performance characteristic curves of the KITACO NS-100 centrifugal pump coupled to GOLDEN BOW GP95 diesel engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pump Tested	:	KITACO NS-100 coupled to GOLDEN BOW GP95 diesel engine
Date of Test	:	February 16, 2014
Test Condition		
Ambient Temperature (^{o}C)	:	27.3
Atmospheric Pressure (mb)	:	1013
Pumping Water Temperature (^{o}C)	:	24.0

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
2478	2091	0.47	9.40	33.9	16.5
2495	2102	1.84	8.03	31.9	16.2
2492	2101	3.22	6.65	30.7	15.5
2494	2108	4.58	5.29	29.0	15.4
2518	2134	5.99	3.88	27.3	13.1
2546	2174	7.41	2.46	25.7	9.74
2549	2186	8.85	1.03	25.0	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

Where: $$H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted to meter.

Computed by determining the velocity of liquid on the pipe by formula, V = Q/A where V, velocity of liquid inside the pipe in meter per second, Q, Discharge rate of the pump in liters per second, and A, Cross-sectional area of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power, WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power, IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters
P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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N.B.

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